

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Flow Assurance Analysis Report
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-016
Document Revision	R01
Document Status	Issued for Review
Originator	Basil AKPAN
Security Classification	Restricted
Issue Date	7-Jul-22

Flow Assurance Analysis Report

R01	07.07.22	Issued for Review	B. Akpan	O.Ezeagu	A.Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	07-07-22	Issued for Review
R02		
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-016

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum

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Definition of Terms

TERMS	DEFINITIONS
BFPCL	Brass Fertilizer & Petrochemical Limited.
SPDC	Shell Petroleum Development Company Nig. Limited
Employer	Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Measuring Range	Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).
On-Line Instruments	Instruments connected to process and utility lines or equipment via primary isolation valves.
Project	Front End Engineering Design for Gas Gathering Pipelines (Phase 1)

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a greenfield methanol production facility in Akabel (Odioma), Brass LGA Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields, namely: Bubouwe Bou & Elepa, in OML 33, Nun River, Seibou, in OML 35, and Opugbene in OML 36.

The Project involves the development, construction, and operation of a 10,000 metric tonnes per day (MTPD) capacity methanol production plant comprising the following elements:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel, - as shown in Figures 1.1, 1.2, & 1.3.
- Product Export Lines and associated Offshore Single Buoy Mooring (SBM) for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) which are required for the operation of the processing plants.

Tata Consulting Engineers Limited (TCE) is the Project Management Consultant for the project.

The Nodal Gas Gathering Schematics

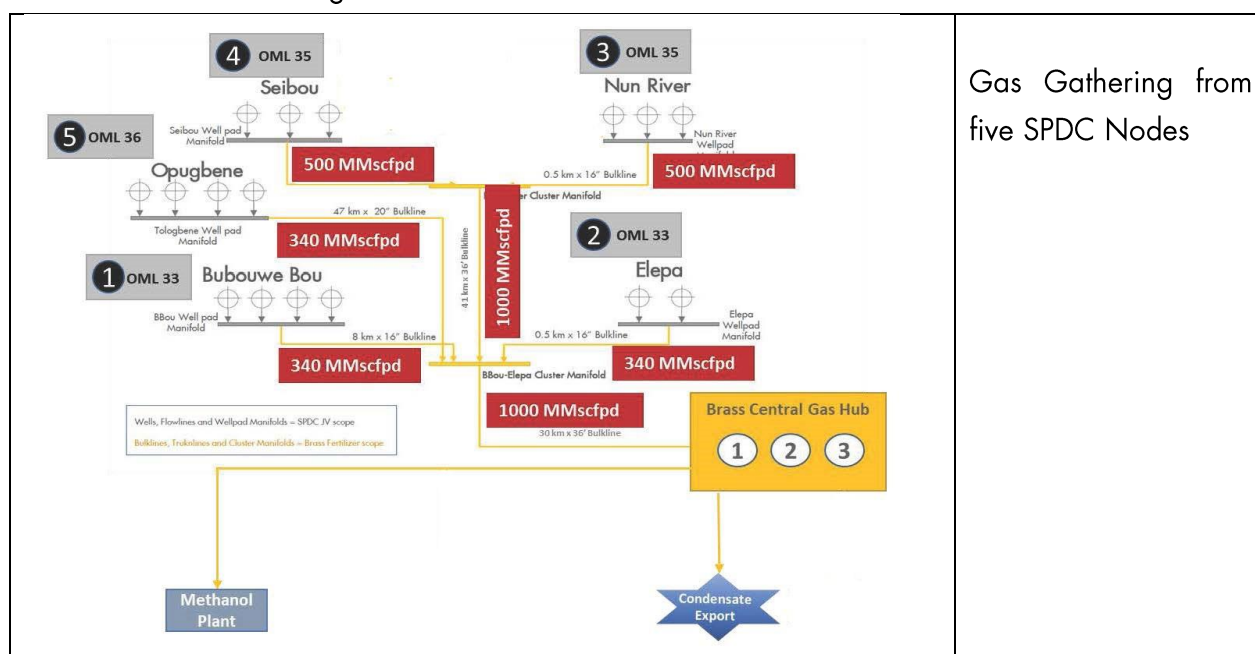
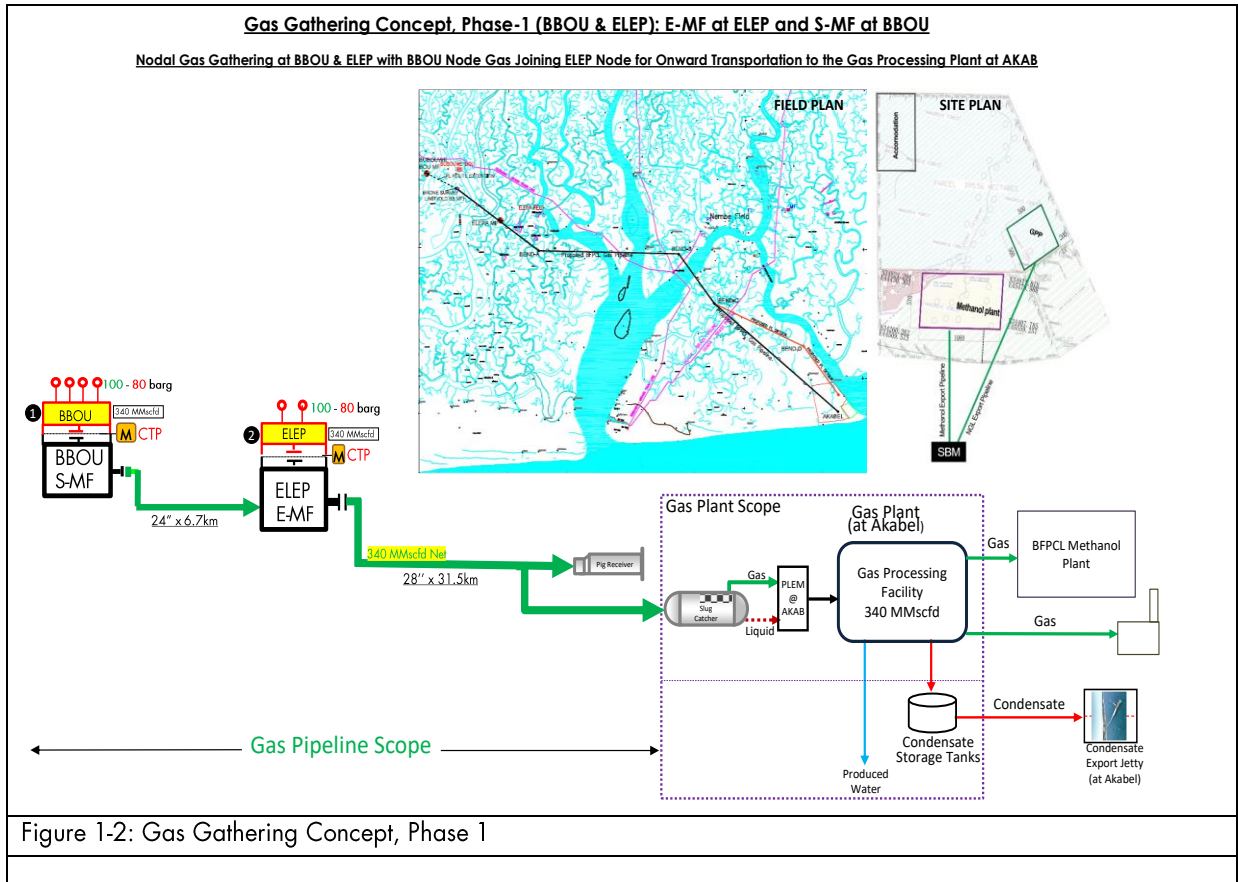
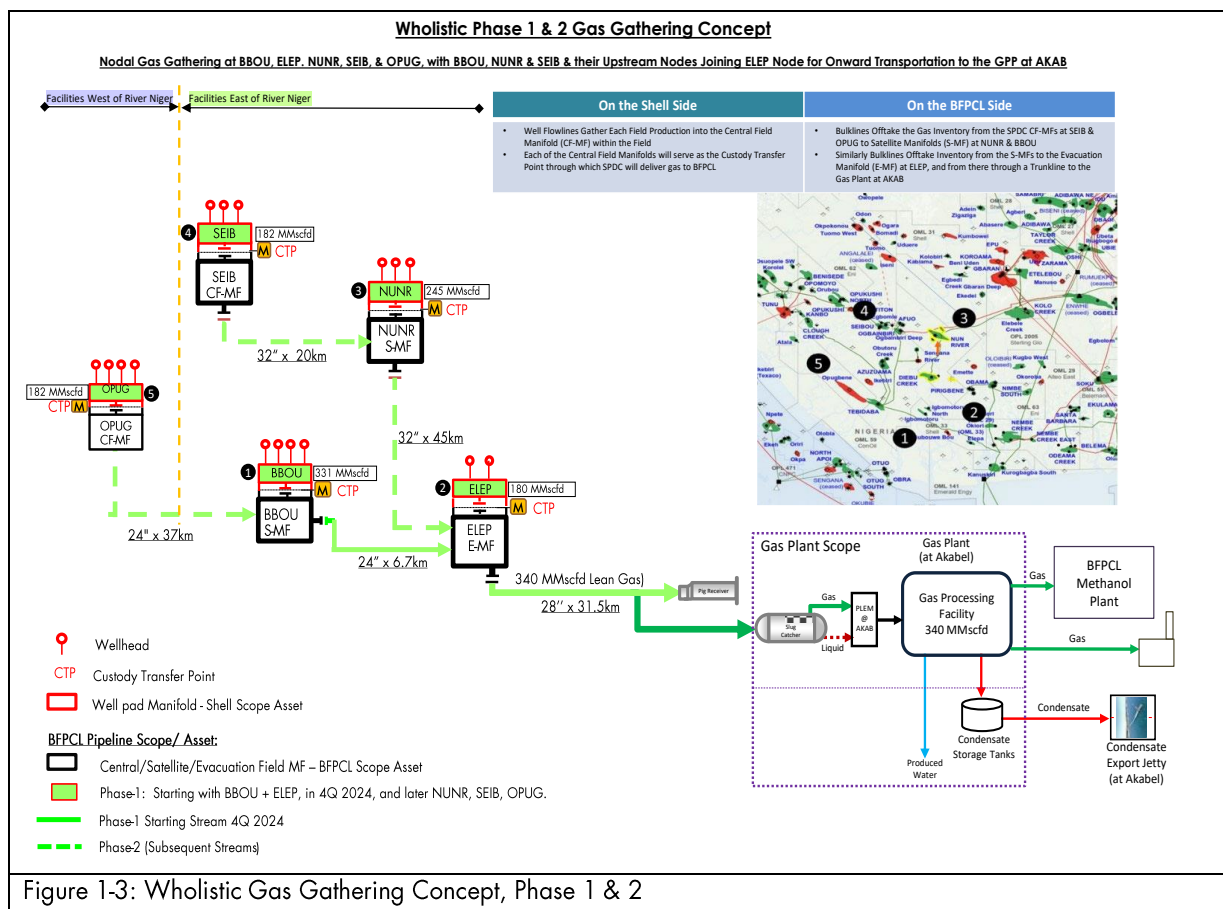


Figure 1-1: Field & Nodes covered in the Gas Gathering

The Nodal Gas Gathering Schematics (contd.)





1.2. Purpose of Document

The purpose of this document is to provide the technical basis for the development of Project Philosophies and Specifications, including all the necessary information, design data, codes and standards required for the FEED of the gas pipelines and associated facilities.

The design and construction of the pipelines shall fully comply with all relevant contractual requirements specified in the Scope of Work and Client's Standards.

The gas gathering will be in two phases, as shown in the schematics below:

- a. Phase 1 comprising gas gathering from Bubouwe Bou and Elepa fields, and
- b. Phase 2 comprising gathering from Nun River, Seibou, and Opugbene fields.

1.2.1. The Phase 1 Facilities

The phase 1 facilities (shown in Figure 1.1) are:

- BBOU – ELEP pipeline
- ELEP - AKAB Pipeline
- BBOU satellite Manifold
- ELEP evacuation Manifold, and

- The Pipeline End Manifold at AKAB.

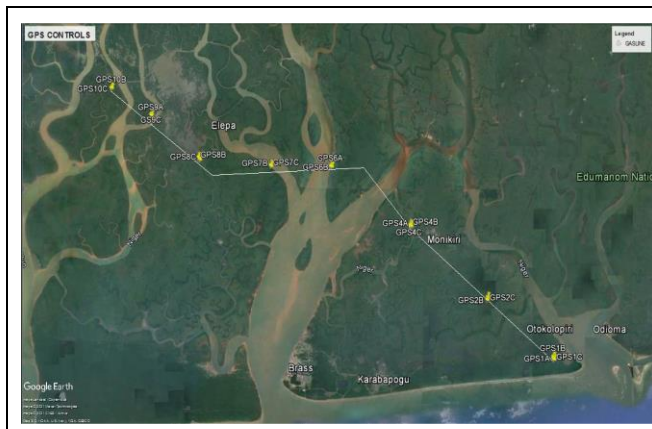
These phase 1 facilities are the focus of the present FEED, however with consideration for provisions and retrofits that will accommodate the phase 2 facilities in a rather seamless manner.

The pipeline route proposed for this phase 1 has been surveyed, and the survey report issued for the FEED. The phase 2 pipelines routes are yet to be defined and surveyed.

1.2.2. The Phase 2 Facilities

The phase 2 facilities comprise NUNR, SEIB, and OPUG Manifolds with their pipelines connecting to phase 1 facilities as shown in Figure 1.2.

1.3. The Phase 1 Pipelines Route / Right of Way



The pipeline route / Right of Way, running from BBOU through ELEP to AKAB is shown in the satellite image below

Figure 1-3: Satellite Image of the proposed BBOU – ELEP-AKAB Right of Way

1.4. Scope

The Facilities to be covered under project scope are:

- Satellite Manifold at BBOU complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from OPUG.
- Evacuation Manifold at ELEP complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from NUNR.
- Pipeline from BBOU to ELEP
- Pipeline from ELEP to AKAB (up to Inlet Flange of Slug Catcher)
- PIG Launchers at BBOU and ELEP Manifold
- PIG Receivers at ELEP and AKAB
- Manifold control system also to be provided. Control Room required at each Manifold to house the Manifold Control system & Telecom system
- Cabinet Room to house the electrical MCC.

- UPS Power for Control System, associated Civil, Mechanical, Pipeline, Electrical, Instrumentation & Control, Telecommunication, and all other supporting infrastructure.

The Scope of the FEED Engineering covers the provision of all necessary information, technical drawings, philosophies, specifications, design data, listing of codes and standards required for the detailed design and construction of the Multiphase pipeline system.

1.5. Design Battery limits

The battery limits for this Phase 1 FEED Project are:

1.5.1. BBOU Platform

- The terminal flanged ball valve on the manifold collecting well fluids from BBOU Wells.
- The flanged ball valve provided for the tie-in of Opugbene (OPUG) Pipeline on the manifold transporting well fluids to ELEP.

1.5.2. ELEP Platform

- The terminal flanged ball valve on the manifold collecting well fluids from ELEPA Wells.
- The flanged ball valve provided for the tie-in of NUN River Pipeline on the manifold transporting well fluids to AKAB.

1.5.3. Akabel Facility

- The Pipeline terminal flange at the inlet to the slug catcher at AKAB.
- The drain outlet flange on the pig receiver at AKAB.

2. Regulations, Codes And Standards

Document Number	Title
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-001- 3	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/MAT-RPT-001	Pipeline Materials Selection & Corrosion Analysis Study
EA/SPDC-BFPC UGS-GEP/ELE-RPT-001	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-RPT-001	Manifold Platforms Concept & Integration with SPDC
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Gas Processing Technology Option
6064/PLRS/001a-PRP	Onshore Pipeline Route Survey for BFCL; Topographical Surveys
6064/PLRS/001PRP	Onshore Pipeline Route Survey for BFCL; Soil Investigation Surveys
40km Gas Supply Pipeline Project ESIA	Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited - Draft Report November 2020.

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Flow Assurance Analysis

3.1. Purpose

This report presents the flow assurance analysis of the Multiphase Production Pipelines from BBOU to Akabel. The analysis comprises the following:

- Steady State Analysis – to determine operating conditions over field life, confirm/validate the pipeline size against the steady state hydraulic analysis results.
- Slugging Assessment – to determine the impact of the line size selected on slugging and pipeline operability by reporting the flow regime and the instantaneous liquid arrival at Akabel.
- Pigging Analysis – to determine the liquid surge volume as a result of pigging operation to aid the design of slug catcher.
- Cooldown and Pipeline Depressurisation Analysis – to determine the cooldown rate and assess any risk of hydrate when the pipeline is shut-in for extended periods as well as during depressurisation of the pipeline:
 - Liquid arrival volume at Akabel
 - Minimum fluid temperature downstream of the RO.
 - Peak gas flowrate
- Hydrate Risk Assessment – to assess the risk of wax or hydrate formation during steady state and transient operation.

In addition to the results of the purpose detailed above, this report also presents the data on which the analysis is based and the methodology adopted. OLGA version 2017 is used to perform the analysis.

3.2. Summary of Results

The findings of the analysis are as follows:

3.2.1. Steady State

- The 24 x 28 and 24 x 36 inch pipeline are optimal for delivering the production forecast to Akabel based on the required pressure being within the available pressure range.
- The 24 x 28 and 24 x 36 inch pipeline shows no fluctuation in the pipeline pressure which is representative of stability.
- The erosional velocity limit based on API14E is not exceeded with the 24 x 28 and 24 x 36 inch pipeline so no erosional velocity constraint exists with this line sizes.

3.2.2. Slugging Assessment

- Examination of the flow regimes in the pipelines show that flow is mostly stratified along the pipeline except at low point where it is slug flow.
- Instantaneous liquid arrival flowrates at Akabel however show significant fluctuations for the 24 x 36 inch pipeline. This translates into significant levels of fluctuation in liquid arrival at Akabel. In contrast, stable liquid arrival flowrate is obtained for the 24 x 28 inch pipeline.

3.2.3. Turndown Assessment

- When the total flowrate is lower than 150 MMscfd the system is gravity dominated while when the total flowrate is higher than 150 MMscfd the system is friction dominated.
- The fluid velocity for flowrates lower than 150 MMscfd is less than the minimum specified by API 14E (flow velocity should not be less than 3 ft/s to minimize deposition of sand and other solids).

3.2.4. Hydrate Assessment

- At the minimum swamp temperature of 18°C, there is no risk of hydrates forming in the pipeline during steady state operation as the steady state temperature is above the hydrate formation temperature at the operation pressure of 74 bara. Also, at the settle-out pressure of 66 bara during pipeline shutdown, it is concluded that there is no risk of hydrate formation in the pipeline during shutdown conditions.

3.2.5. Cooldown & Pipeline Depressurization Results

- The pipeline temperature profile during steady state operation shows the fluid temperature is above the hydrate formation temperature at the pipeline outlet/arrival at Akabel. Hence cooling of the pipeline following shutdown poses no additional hydrates risk.
- On pipeline shutdown, the pipeline settles-out at approximately 66 bara. At this pressure and ambient temperature (minimum swamp temperature of 18°C) no hydrate risk exists.
- The rate of pipeline depressurization is dependent on the RO size. The optimum RO size however cannot be determined as further analysis to include the pipework downstream of the RO should be considered and the impact of the depressurization rate, RO size and subsequent Joule-Thompson cooling on minimum wall temperature assessed to inform the RO size. This is recommended to be done at the detailed engineering design stage.
- The lowest temperature obtained downstream the RO (4-inch) is -16.9°C during pipeline depressurization.
- The water outflows from the pipeline during depressurization is negligible for the RO sizes

considered.

- The oil outflow from the pipeline increases with RO size increase amounting to 487m³ for RO size of 4-inch.
- The maximum/peak instantaneous gas flowrate of 334MMscf/d is obtained during pipeline depressurization when an RO of 4 inch is used.

3.2.6. Pigging Results

- The pig residence time when travelling at an average velocity of 1 and 1.5m/s is approximately 8 and 5.5 hours respectively.
- During pigging operation, a maximum accumulated volume of 3500 m³ is calculated.

4. Flow Assurance Basis

The basis of the flow assurance analysis is presented as follows and is primarily a compilation of the data extracted from the project basis of design [Ref Error! Reference source not found.], the Hydraulic Analysis report and the pipeline survey report. The reference for each data source is however still given within the narrative.

4.1. Manifold Conditions

The manifold conditions for the various fields are presented in Table 1 below. This is based on currently available data. The lowest manifold temperature of 50°C is used for the hydraulic and hydrate assessment.

Table 1 – Manifold Conditions

S/N	Field	Manifold Flowing Temperature (°C)	Manifold Flowing Pressure (barg)
1	BBOU	50 – 95	80 – 100
2	ELEPA	50 – 95	80 – 100
3	OPUG	50 – 95	80 – 100
4	SEIB	50 – 95	80 – 100
5	NUNR	50 – 95	80 – 100

4.2. Pipeline Inlet Fluid Composition

The fluid composition shown below is extracted from the Project Design Basis document.

Field 1: Bubouwe Bou Analogue		Field 2: Elepa		Field 3: Nun River H4.0	
Component	Mol.%	Component	Mol.%	Component	Mol.%
N ₂	0.07	N ₂	0	N ₂	NA
CO ₂	0.59	CO ₂	2.22	CO ₂	0.77
C ₁	92.92	C ₁	81.66	C ₁	72.83
C ₂	3.16	C ₂	6.13	C ₂	7.68
C ₃	1.01	C ₃	2.59	C ₃	4.85
iC ₄	0.61	iC ₄	0.75	iC ₄	1.46
nC ₄	0.44	nC ₄	0.81	nC ₄	1.97
iC ₅	0.25	iC ₅	0.45	iC ₅	0.95
nC ₅	0.19	nC ₅	0.33	nC ₅	0.76
C ₆	0.23	C ₆	0.65	C ₆	1.03
C ₇	0.19	C ₇	0.9	C ₇	1.41
C ₈	0.04	C ₈	0.87	C ₈	1.61
C ₉	0.03	C ₉	0.38	C ₉	0.9
C ₁₀	0.02	C ₁₀	0.3	C ₁₀	0.65
C ₁₁	0.03	C ₁₁	0.23	C ₁₁	0.47
C ₁₂₊	0.22	C ₁₂	0.22	C ₁₂	0.37
C ₁₃	NA	C ₁₃	0.23	C ₁₃	0.34
C ₁₄	NA	C ₁₄	0.22	C ₁₄	0.3
C ₁₅	NA	C ₁₅	0.19	C ₁₅	0.26
C ₁₆	NA	C ₁₆	0.15	C ₁₆	0.21
C ₁₇	NA	C ₁₇	0.12	C ₁₇	0.16
C ₁₈	NA	C ₁₈	0.11	C ₁₈	0.14
C ₁₉	NA	C ₁₉	0.06	C ₁₉	0.08
C ₂₀₊	NA	C ₂₀₊	0.43	C ₂₀₊	0.8
TOTAL	100	TOTAL	100	TOTAL	100
Flash CGR	7	Flash CGR	59.7	Flash CGR	161.7

4.3. Hydrate Dissociation Curve & Phase Envelope

Figure 1 below shows the hydrate dissociation curve for Boubowe, Elepa and combined fluid (Opughene and Nun River considered in the combined fluid). Please note that no margin was applied and the saturated water is unsalted.

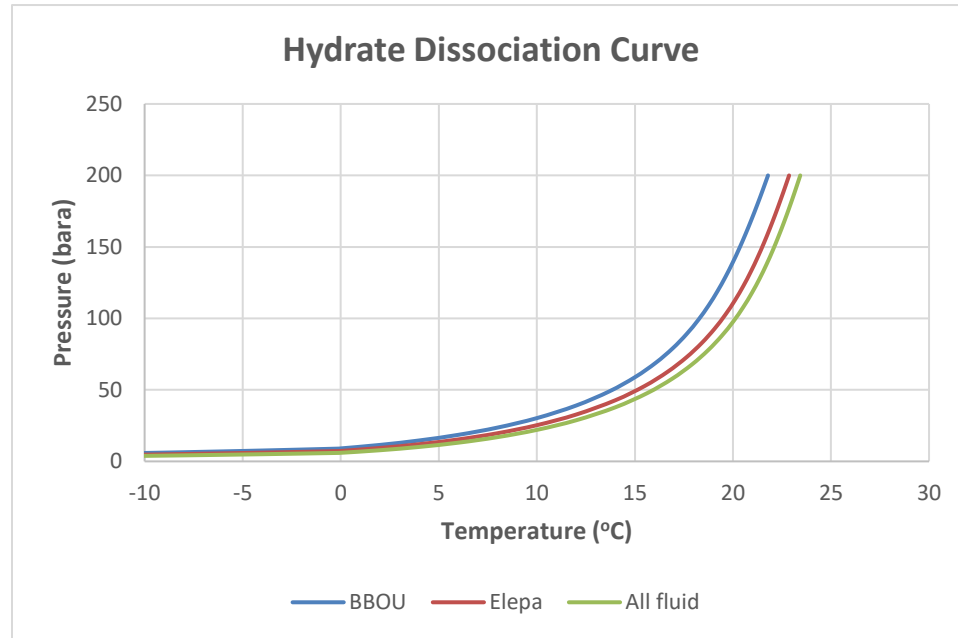


Figure 1 - Hydrate Dissociation Curve_Unsalted

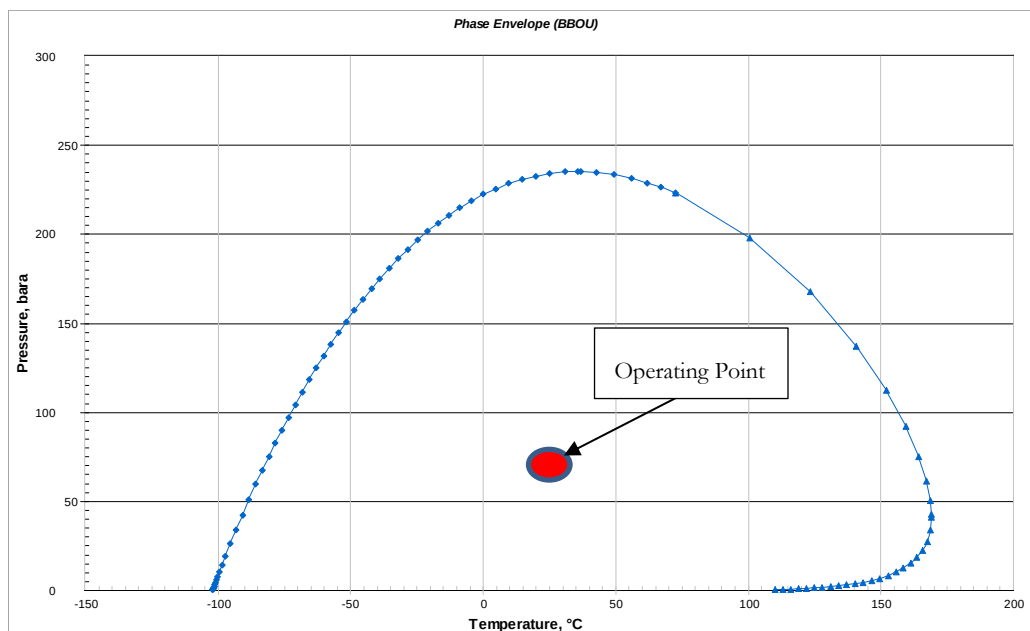
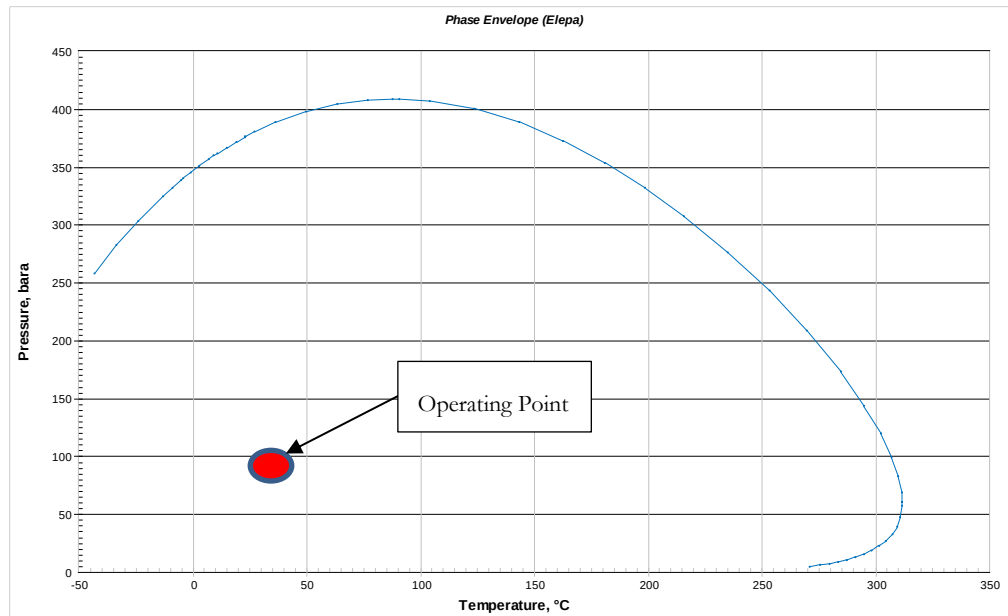


Figure 2 – Phase Envelope for BBOU Fluid*Figure 3 - Phase Envelope for Elepa Fluid*

4.4. Pipeline Configuration

The details of the pipeline topography are presented below.

4.4.1. Pipeline Topography

The proposed pipeline topography is extracted from the survey report of the pipeline corridor (the detailed routing data provided in Microsoft Excel). The topography was provided with data points at every 25 meters. This was simplified to improve simulation runtime but still ensuring low points in the pipeline where liquid accumulation will occur is captured to allow accurate quantification of the net forward flow of the liquid phase. The pipeline topography is presented below.

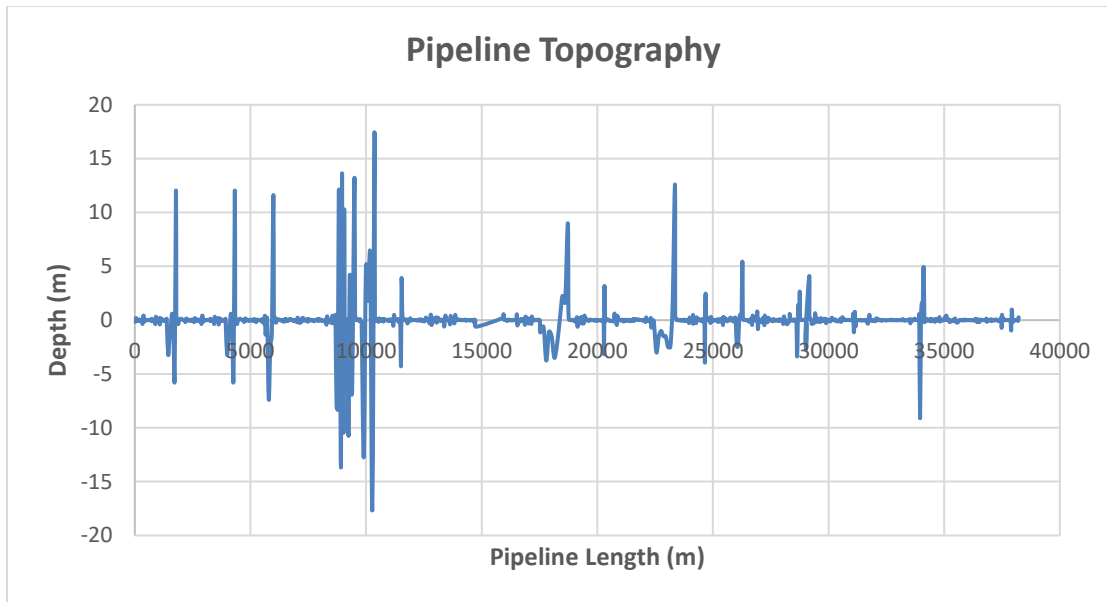


Figure 4 – Bobouwe to Akabel Pipeline Topography

4.4.2. Pipeline Material Characteristics

The pipeline wall thickness and coating data used for the analysis are presented below.

Table 2 – Pipeline Sizes Considered & Coatings

Wall Section	Material	Outer Diameter (inch)	Thickness (mm)	Inside Diameter (mm)
Pipeline	24-inch Carbon Steel	24	15.88	578.24
	28-inch Carbon Steel	28	17.48	676.04
	36-inch Carbon Steel	36	23.83	866.34
Layer-1	3LPE		3.2*	
Layer-2	Concrete		40	

*Information is assumed from previous project

A roughness of 45 micro meters is used for the analysis.

4.4.3. Pipeline Material Properties

The material properties used in the analysis are presented below

Table 3 – Pipeline Material Properties

MATERIAL	THERMAL CONDUCTIVITY (W/mk)	HEAT CAPACITY (J/kgK)	DENSITY (Kg/m ³)
Carbon Steel	45	470	7850
3LPE	0.3	2200	900
Concrete	1.8	880	2250

The burial depth of the pipeline is assumed to be 1.5 meters

4.5. Ambient Temperature Conditions

The ambient temperature conditions are presented in the table below

Table 4 – Ambient Temperature Conditions

	Units	Value
Minimum ambient temperature	°C	18
Minimum Soil temperature	°C	23
Minimum swamp temperature	°C	23

4.6. Arrival Pressure at Akabel

The arrival pressure boundary condition at Akabel is 65 bara.

4.7. Production Profile

See below production forecast for periods considered (minimum gas production and Maximum gas production/maximum liquid production).

Table 5 – Production Forecast for Selected Periods

Year	Gas (MMSCFD)	Condensate (BCPD)	Water (BWPD)
9	356	2724	9275
14	386	22905	16513

Produced water composition was not available and therefore the fluid was saturate with water. It is recommended that a detailed production forecast with produced water composition should be available for further studies in detailed engineering

5. Methodology

5.1. Steady State

The steady state line size is assessed based on the following:

- Available versus require pressure at the WHP manifold
- Erosional velocity ratio based on API 14E
- Pipeline stability/slugging

The methodology adopted is to simulate the flow operation for a range of pipeline sizes. Results are only presented for a 24 x 28 and 24 x 36-inch pipeline as selected from the steady state hydraulic report.

OLGA 2017 is used to simulate the flow conditions in the multiphase flow pipeline for the selected periods in field-life. A fixed arrival pressure is defined, 65 bara, and the phases flowrate as defined by the period in field life are specified as a mass source. The back pressure at Elepa and Bobouwe is calculated.

5.2. Slugging & Pigging Analysis

The methodology adopted for the slugging determine the maximum surge volume during steady state production (slugging surge volume) and during pigging/pig arrival (pigging surge volume). In essence the surge volume is the 'positive volume delta' in the separator that will result due to a combination of liquid slug inflow to the slug catcher/separator from the pipeline and outflow of liquid at the maximum slug catcher/separator drain rate.

The pigging velocities considered are 1 and 1.5 m/s.

The pig is inserted after 3 hours of simulation at the steady state production flowrate. This is to ensure steady state conditions particularly the liquid holdup, is achieved prior to launching the pig. The production rate is ramped-down to a predetermined flowrate required to maintain the pig at an average velocity of 1 and 1.5m/s. The simulation runs are typically run for 10 hours.

The analysis assumes a default pig details in OLGA. This is a low resistance pig resulting in low pressure drop across the pig and hence have negligible impact on the simulation results which is primarily to determine the volume of liquid slug arriving at Akabel.

5.3. Cooldown & Depressurization Analysis

The pipeline blowdown is simulated for the maximum flowrate operating case and comprises steady state operation followed by shutin of the pipeline (cooldown) and depressurizing the pipeline via a restriction orifice. The pipeline is allowed to cool to ambient temperature prior to depressurizing it. A leak model in OLGA which has a leak diameter (RO size) and back-pressure

specified is used to define the relief rate. The depressurization analysis is performed for a range of RO sizes.

5.4. Hydrate Risk Assessment

From the steady state and transient simulation results based on 'coldest' ambient scenario, the operating pressure and temperature relative to the hydrate formation conditions is used to assess the risk of hydrate formation.

5.5. Turndown Assessment

Steady state analysis will be performed at lower flowrate to determine the minimum flowrate before the onset of instability in the system. The production rate considered were 200 MMscfd, 150 MMscfd, 100 MMscfd and 80 MMscfd.

For each case investigated, the following parameters were monitored:

- Pressure in in the bulkline
- Fluid velocity

In addition, flow instability was checked at the pipeline outlet in order to understand, for each case, if a constant or oscillating flow is expected at the slug catcher.

6. Results

6.1. Steady State

6.1.1. Bulkline

6.1.1.1. 24 x 28 – inch Pipeline

Figure 6 and 6 presents the pressure profile calculated for the 24 x 28-inch pipeline for the maximum and minimum gas cases. The following observations are made:

- The required pressure at Elepa and BBOU is less than the minimum available pressure (80 barg) to deliver the profile flowrates
- The 24 x 28-inch bulkline shows no fluctuation in the pressure and hence is stable and will not result in any operability problems for the entire field life.

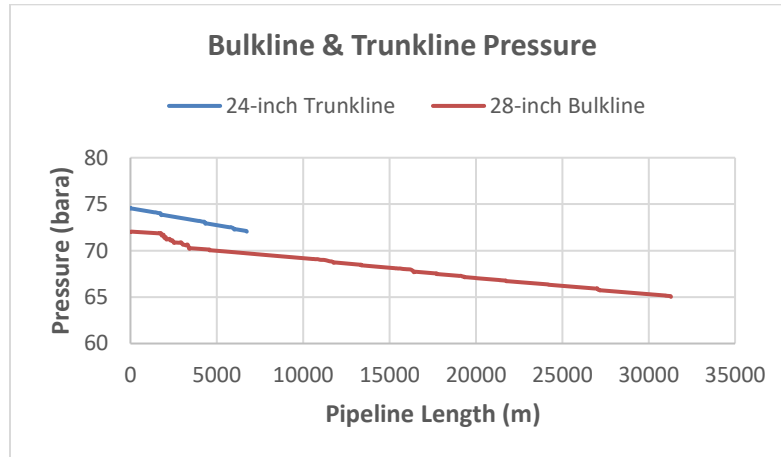


Figure 5 – Pressure Profiles – Maximum Gas Case

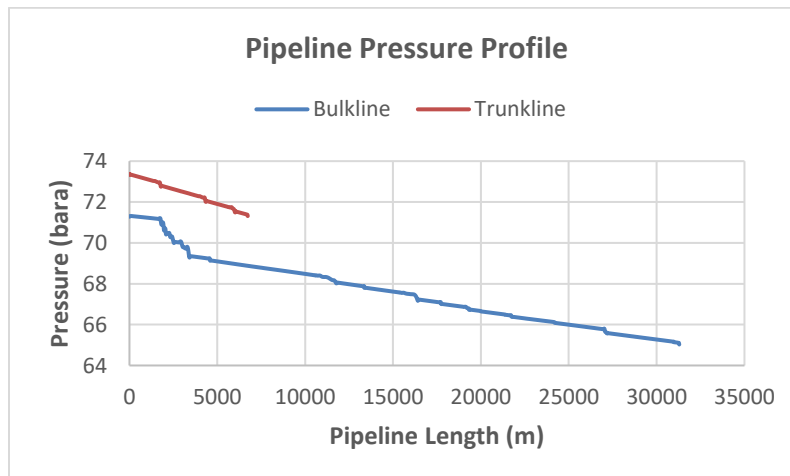


Figure 6 - Pressure Profiles – Minimum Gas Case

Figures 7 and 8 presents the erosional velocity ratio (actual fluid velocity divided by the API14E erosional velocity limit). This shows that the erosional velocity limit is not exceeded with the 24 x 28-inch pipeline so no erosional velocity constraint exists.

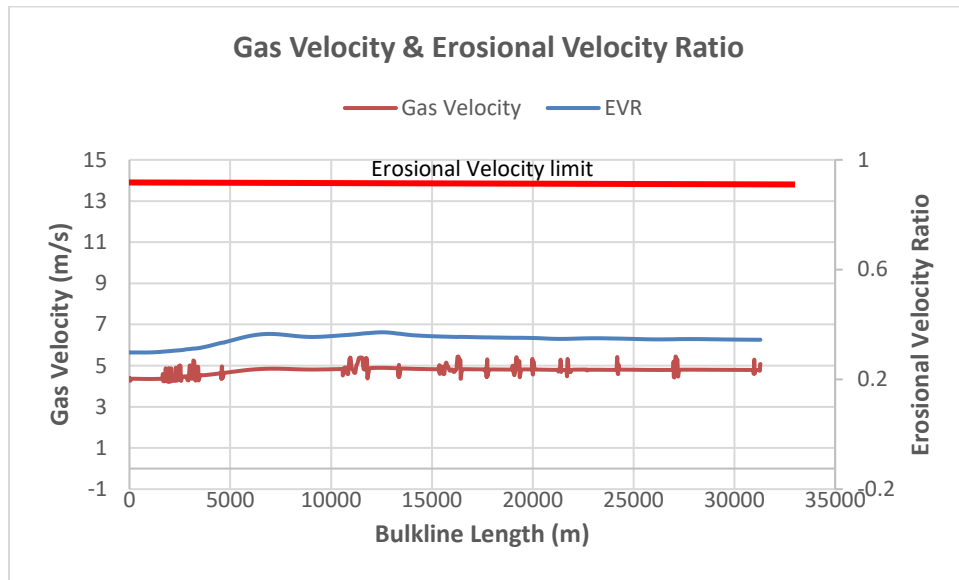


Figure 7 – Gas Velocity & Erosional Velocity Ratio – Maximum Gas Case

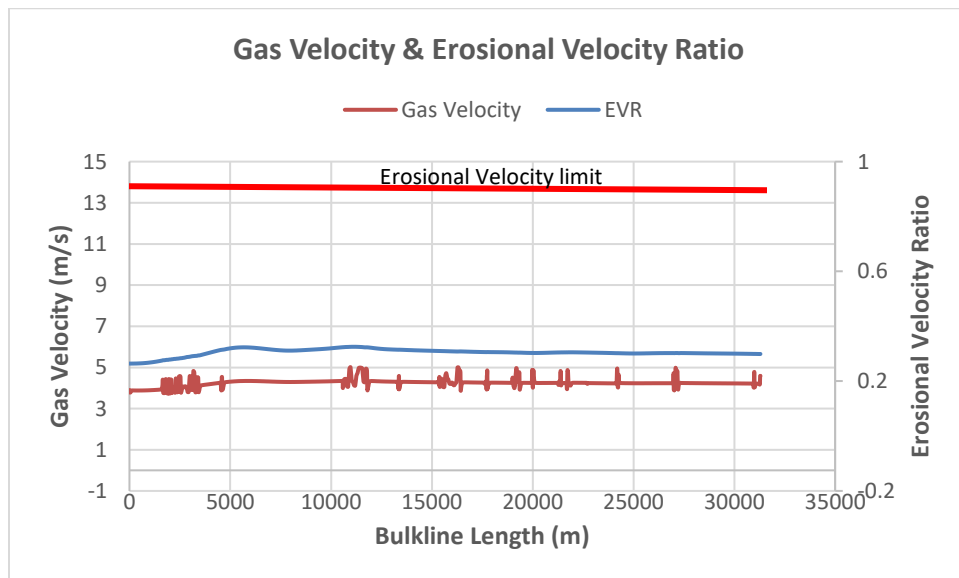


Figure 8 - Gas Velocity & Erosional Velocity Ratio – Minimum Gas Case

6.1.1.2. 36 – inch Pipeline

Figure 9 presents the required inlet pressure calculated for the 36-inch bulkline for the maximum gas cases. The following observations are made:

- The required pressure at Elepa is less than the minimum available to deliver the profile flowrates
- The 36-inch bulkline shows no fluctuation in the pressure and hence is stable

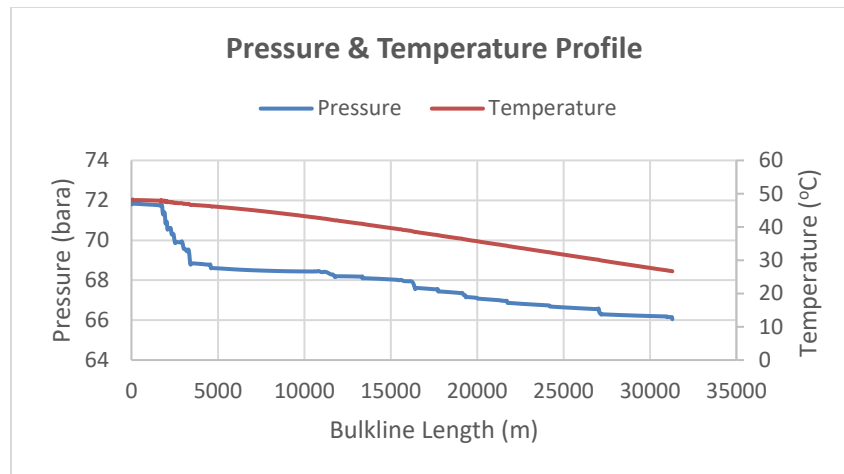


Figure 9 – Pressure and Temperature Profile – Maximum Gas case

Figure 10 presents the erosional velocity ratio (actual fluid velocity divided by the API 14E erosional velocity limit). This shows that the erosional velocity limit is not exceeded with the 24 x 36-inch pipeline so no erosional velocity constraint exists.

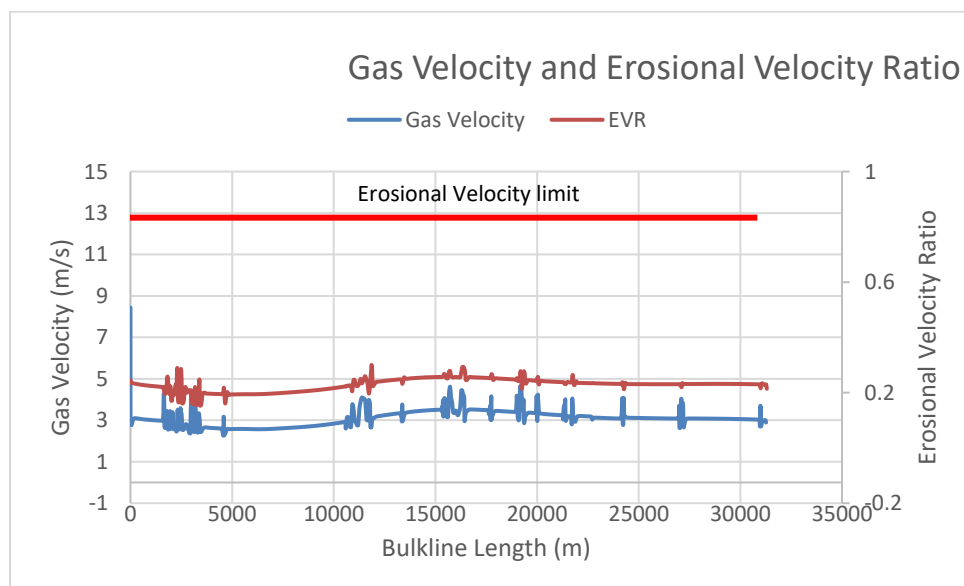


Figure 10 – Gas Velocity & Erosional Velocity Ratio – Maximum Gas Case

6.2. Slugging Assessment

Examination of the flow regimes, during steady state, show mainly stratified flow exists for both 24 x 28 and 24 x 36 inch pipeline sizes. Instantaneous liquid arrival flowrates at Akabel however show significant fluctuations for the 24 x 36-inch pipeline as shown in Figures 11 and Figure 12 which compare the instantaneous liquid arrival at Akabel for both pipeline sizes.

The plots show significant levels of fluctuation in liquid arrival at Akabel when a 36-inch pipeline is selected for the bulkline compared with the relatively stable liquid arrival flowrate when a 28-inch pipeline is selected for the bulkline. However, the operation of the 36-inch pipeline will be challenging from an operation management point of view as this will result in frequent process upset.

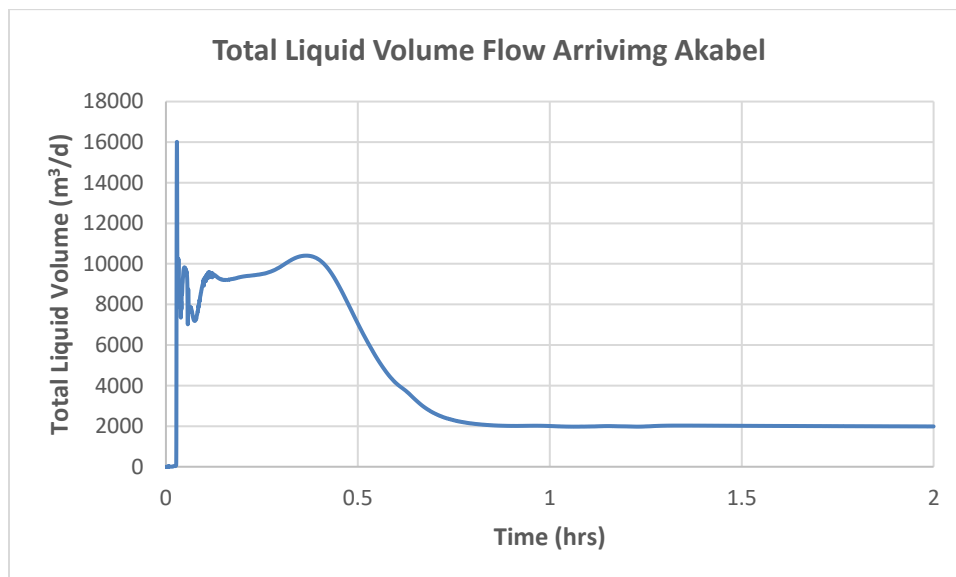


Figure 11 – Instantaneous Liquid Arriving Akabel – Maximum Gas Case_24 x 28 inch

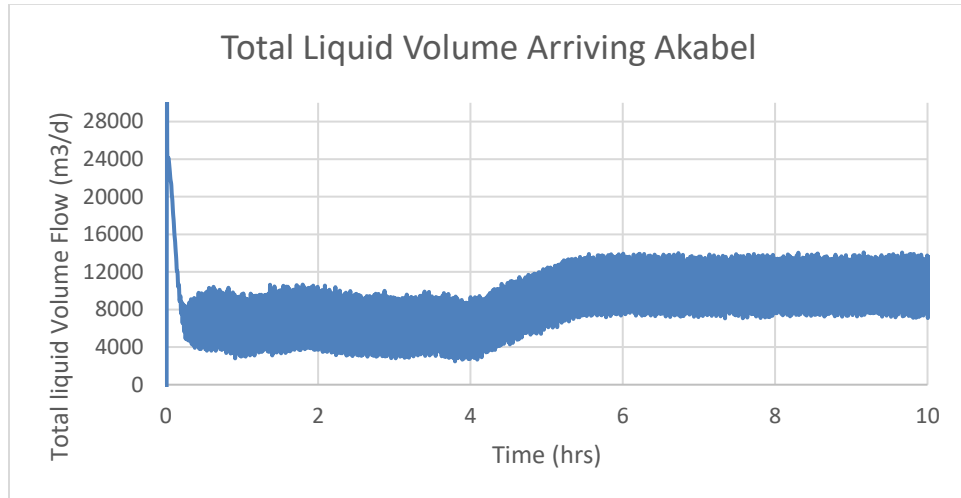


Figure 12 - Instantaneous Liquid Arriving Akabel – Maximum Gas Case_24 x 36 inch

6.3. Hydrate Assessment

There is no risk of hydrate formation at steady state or normal shutdown conditions as the hydrate formation temperature at the operating/settle-out pressure is less than the minimum ambient temperature of 18°C for the 24 x 28 and 24 x 36 inch pipelines.

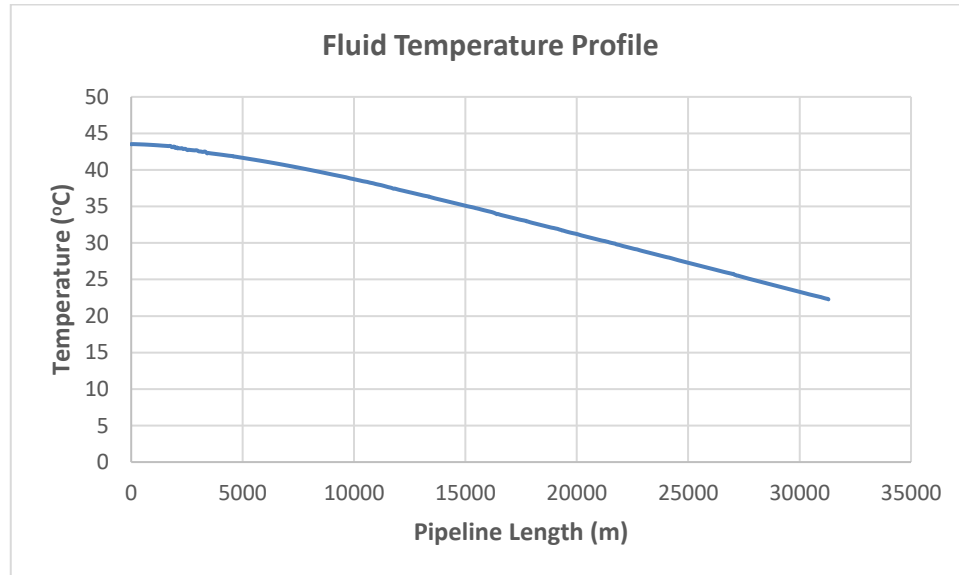


Figure 13 – Bulkline Temperature_ Maximum Gas Case_28-inch

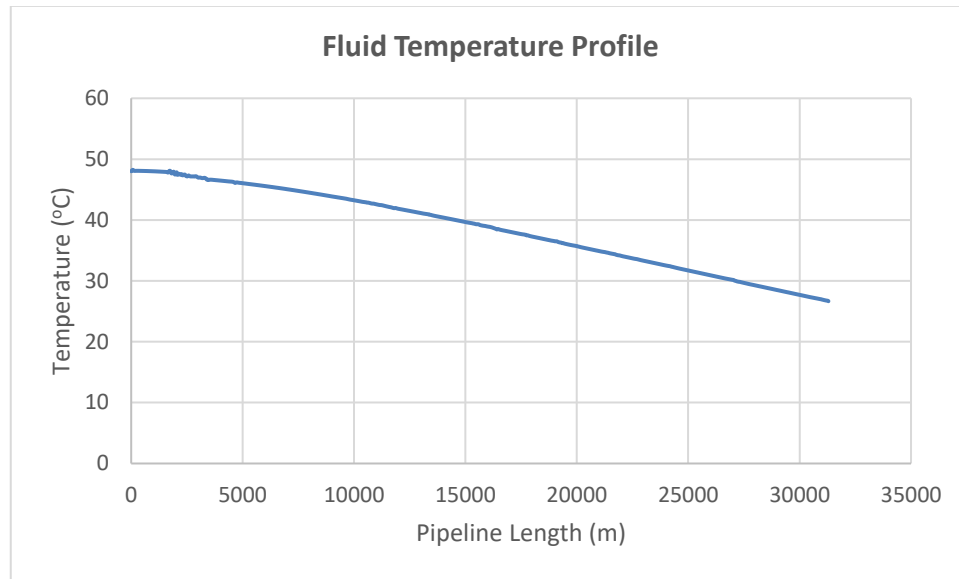


Figure 14 - Bulkline Temperature_ Maximum Gas Case_36-inch

6.4. Turndown Assessment

Figure 15 and 16 below shows the required inlet pressure and fluid velocity respectively for the different flowrate considered. From the results, the following observations were made:

- When the total flowrate is lower than 150 MMscfd the system is gravity dominated while when the total flowrate is higher than 150 MMscfd the system is friction dominated.
- The fluid velocity for flowrates lower than 150 MMscfd is less than the minimum specified by API 14E (flow velocity should not be less than 3 ft/s to minimize deposition of sand and other solids).

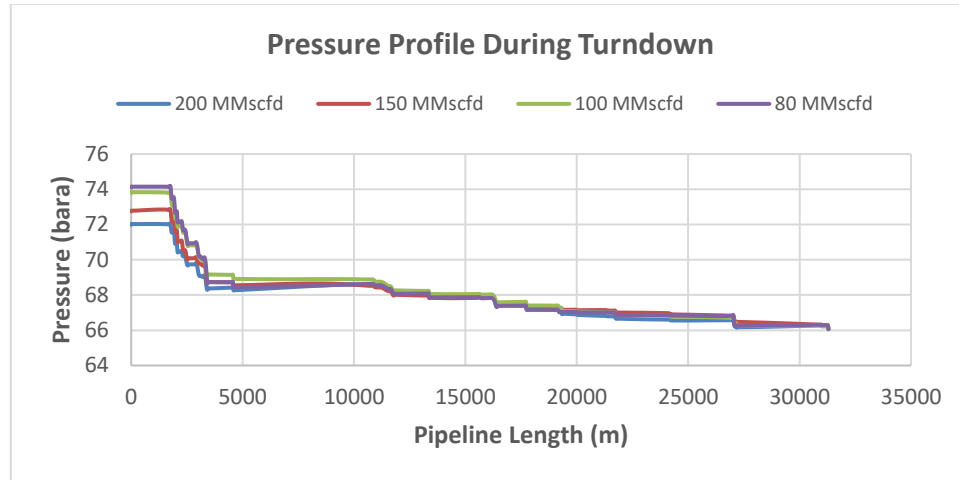


Figure 15 – Pipeline inlet Pressure at Elepa for Different Flowrates During Turndown

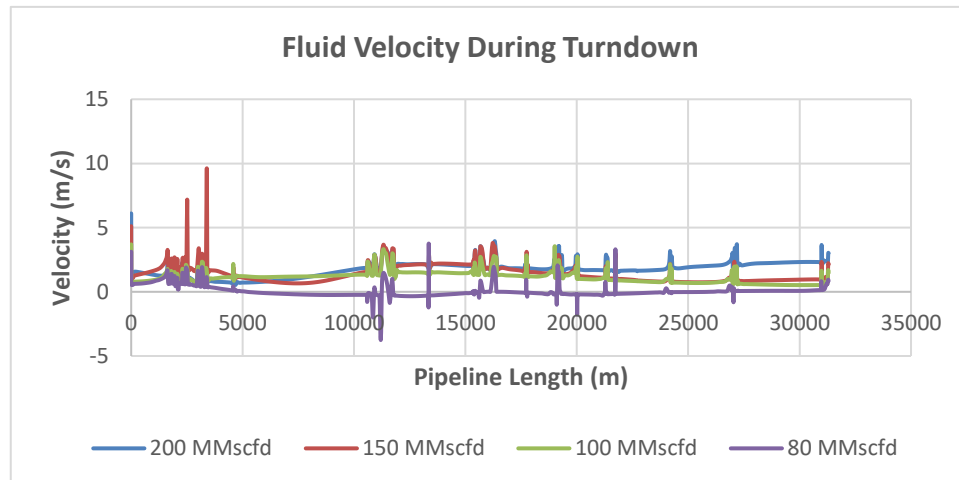


Figure 16 – Fluid Velocity for Different Flowrate During Turndown

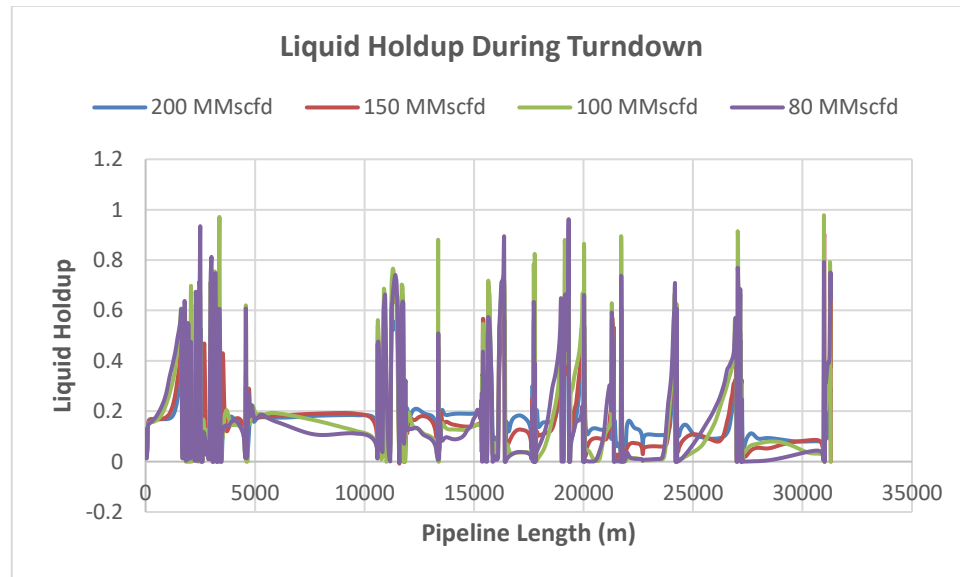


Figure 17 – Liquid Holdup for Different During Turndown

To prevent solid deposition and liquid accumulation at low points in the pipeline it is recommended that fluid flowrate should not be lower than 150MMscfd.

6.5. Cooldown & Depressurization Results

6.5.1. Cooldown Result

There is no risk of hydrate formation after a prolonged shutdown as the ambient temperature is above the hydrate formation temperature of the fluid at the operating pressure of about 70 bara (see Figure 1).

6.5.2. Pipeline Depressurization Results

The result of the pipeline depressurization analysis comprises the following:

1. The pipeline pressure decline rate
2. The fluid temperature upstream of the RO
3. The liquid outflow volume from the pipeline
4. The gas outflow rate from the pipeline

The above is presented for RO sizes of 1, 2, 3 and 4 inches.

6.5.2.1. Pipeline Depressurization Rate

The pipeline depressurization rate for the RO sizes considered is presented in Figure 18 and the depressurization time summarized in Table 6Error! Reference source not found.. The results show t

he pipeline settles-out pressure at approximately 66 bara on shutdown and depressurizes at a rate dictated by the RO size.

The optimum RO size cannot be determined as further analysis to include the pipework downstream of the RO is required and the impact of the depressurization rate, RO size and subsequent Joule-Thompson cooling on minimum wall temperature assessed to inform the RO size. This is recommended to be done at the detailed engineering design stage.

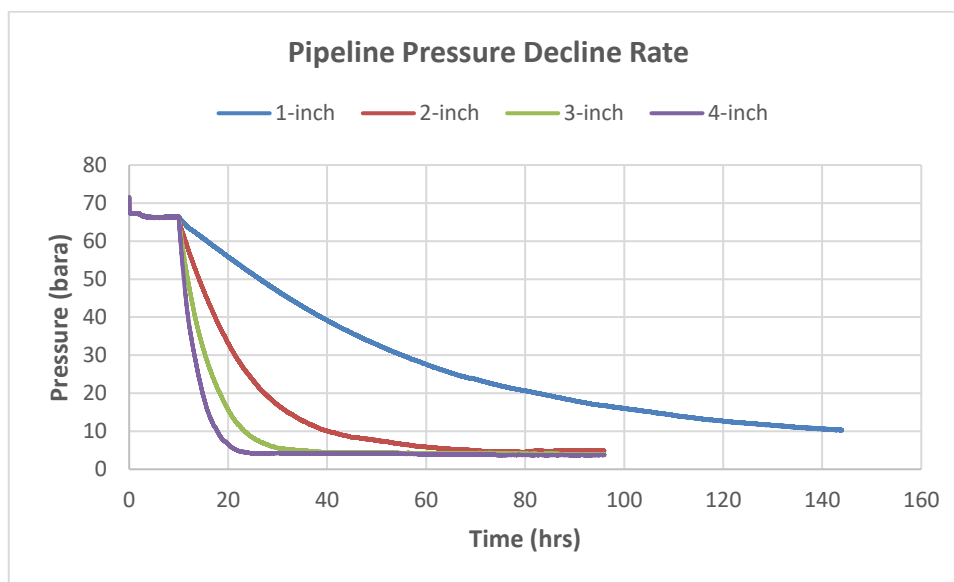


Figure 18 - Pressure Decline rate in the pipeline for different RO

Table 6 – Depressurization Time to 20 and 5 bara

S/N	RO (inch)	Depressurization Time to 20 bara (hrs)	Depressurization Time to 5 bara (hrs)
1	1	72	>140
2	2	17	59
3	3	8	24
4	4	5	12

6.5.2.2. Minimum Fluid Temperature Downstream the RO

The fluid temperatures downstream of the RO is presented Figure 19. This is representative of the lowest temperatures experienced in the pipework downstream of the RO during pipeline depressurization.

Table 7 presents the minimum temperatures obtained for the different RO sizes during depressurization.

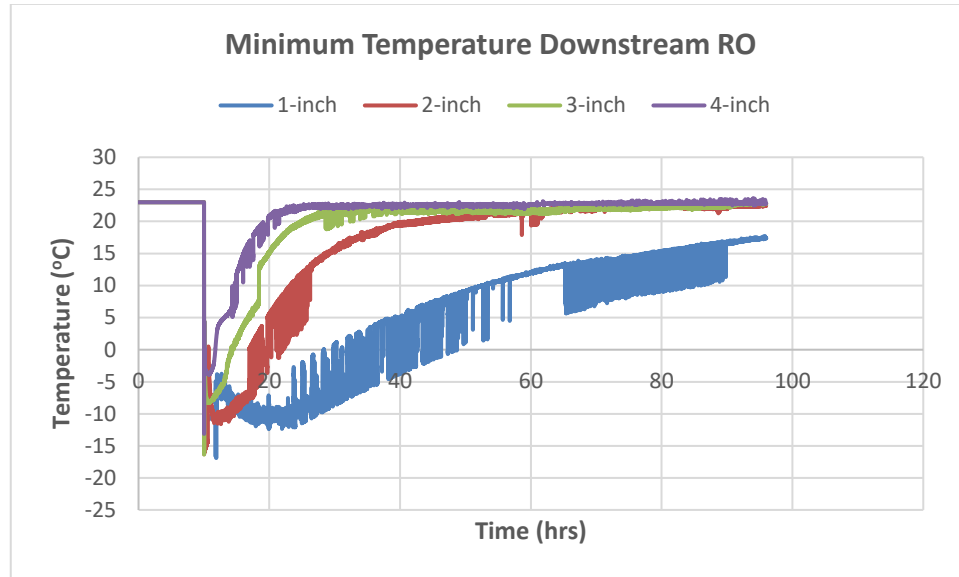


Figure 19 - Minimum Fluid Temperature Downstream the RO

Table 7 – Fluid Minimum Temperature for each RO Size

S/N	RO Size (inch)	Minimum Fluid Temperature (°C)
1	1	-13
2	2	-16.3
3	3	-16.4
4	4	-16.9

6.5.2.3. Liquid Outflow During Depressurization

The total liquid volume outflow from the pipeline during depressurization is determined and presented as follows to inform the flare drum liquid holding capacity. To report this, the accumulated oil and water volumes over the depressurization period are reported for each RO size, the sum of which gives the liquid outflow volume.

Figure 20 and Figure 21 present the plot of the accumulated oil and water volumes during depressurization. It is noted that the 'negative' volumes are the software's convention for representing flow out of the leak point. A summary of the liquid volume outflow is presented in Table 8. The results show the following:

- The water outflows from the pipeline are negligible for the RO cases considered
- The oil volume outflow increases as the RO size increases
- The maximum liquid volume outflow from the multiphase pipeline during depressurisation is about 480 m³.

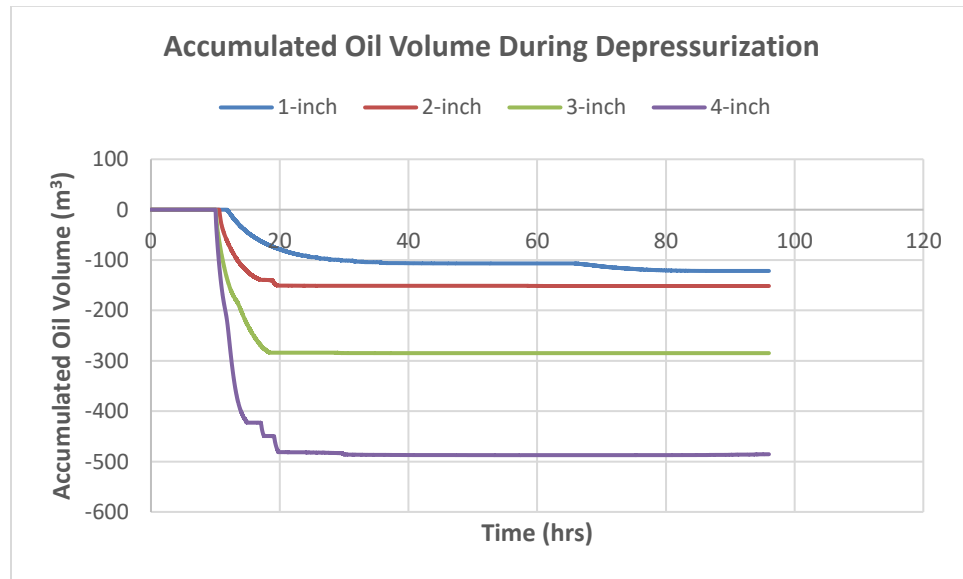


Figure 20 - Accumulated Oil Volume During Depressurization for Varying RO Sizes

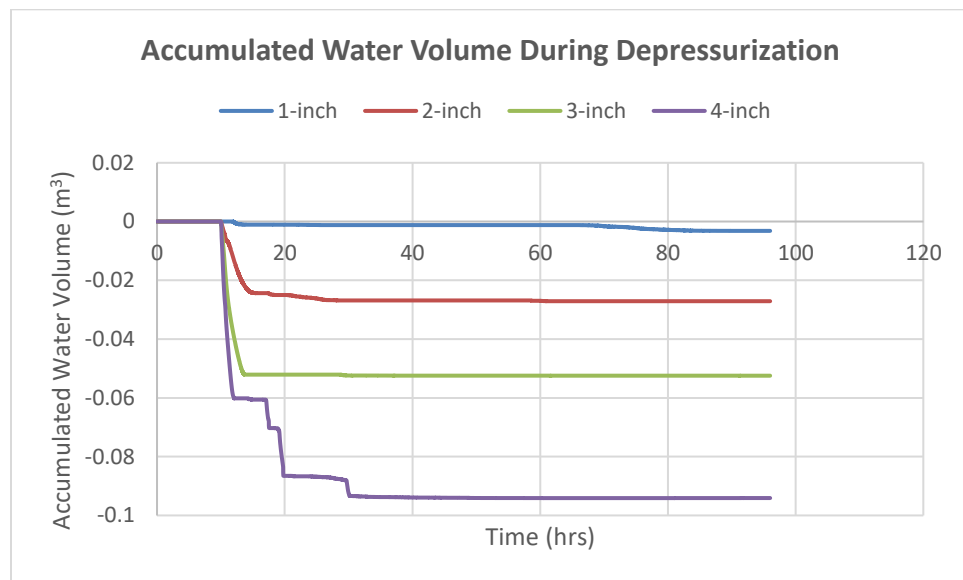


Figure 21 - Accumulated Water Volume During Depressurization for Varying RO Sizes

Table 8 – Total Oil & Water Outflow During Depressurization

S/N	RO (inch)	Oil Outflow Volume (m³)	Water Outflow Volume (m³)
1	1	119	0.003
2	2	151	0.027
3	3	285	0.052
4	4	487	0.094

6.5.2.4. Gas Outflow During Depressurization

The gas flowrate through the RO during depressurization is presented in Figure 22. The plot shows the increasing gas flowrate with RO size and a maximum/peak instantaneous gas flowrate of 334 MMscf/d when an RO of 4 inch is used. Table 9 summarizes the maximum gas flowrates for each RO size.

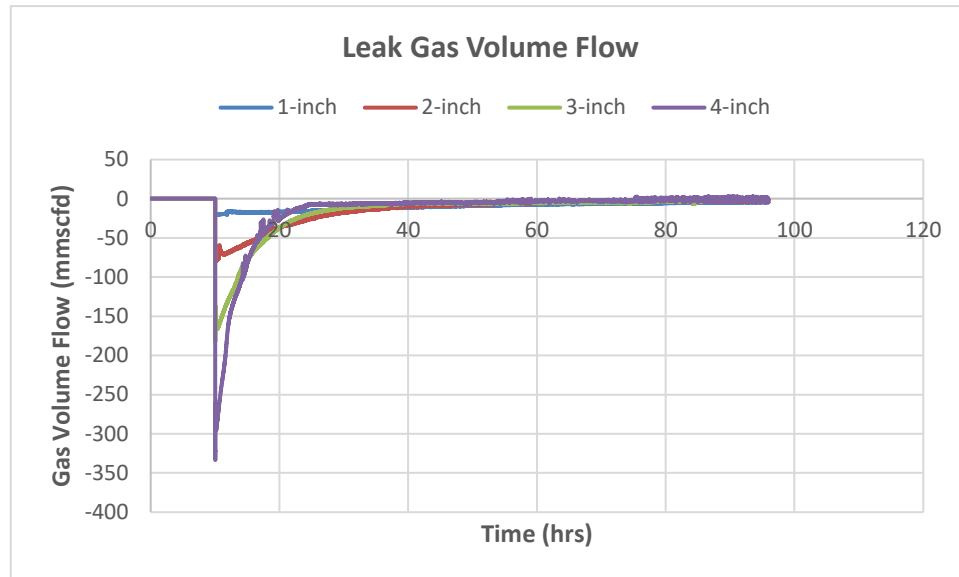


Figure 22 - Gas Flowrate During Depressurization for Varying RO Size

Table 9 – Peak Gas Flowrate During Depressurization

S/N	RO Size (inch)	Instantaneous Gas Volume Flowrate (MMSCFD)
1	1	21
2	2	81
3	3	182
4	4	334

6.6. Pigging Results

The results of the pigging analysis are presented as follows based on the 36-inch multiphase pipeline selection. The results comprise the following:

- The production flowrate needed to achieve an average pig velocity of 1 and 1.5m/s
- The surge volumes i.e., the volume increase that will occur due to the sweeping of the liquids in the pipeline during the pigging operation.

Table 10 presents the liquid flowrate (estimate) required to achieve average pig velocities of 1 and 1.5m/s. Figure 23 and Figure 24 present the average pig velocity achieved by these flowrates. The pig location/travel distance is also shown. The plots show the pig residence time as approximately 8 and 5.5 hours for a pig velocity of 1 and 1.5m/s respectively. For each plot presented in this section, the pig is launched at 'Time [hr] = 3'.

Table 10 - Flowrate required to Achieve Average Pig Velocities of 1 & 1.5 m/s

S/N	Pig Velocity (m/s)	Required Flowrate (MMSCFD)
1	1	140
2	1.5	210

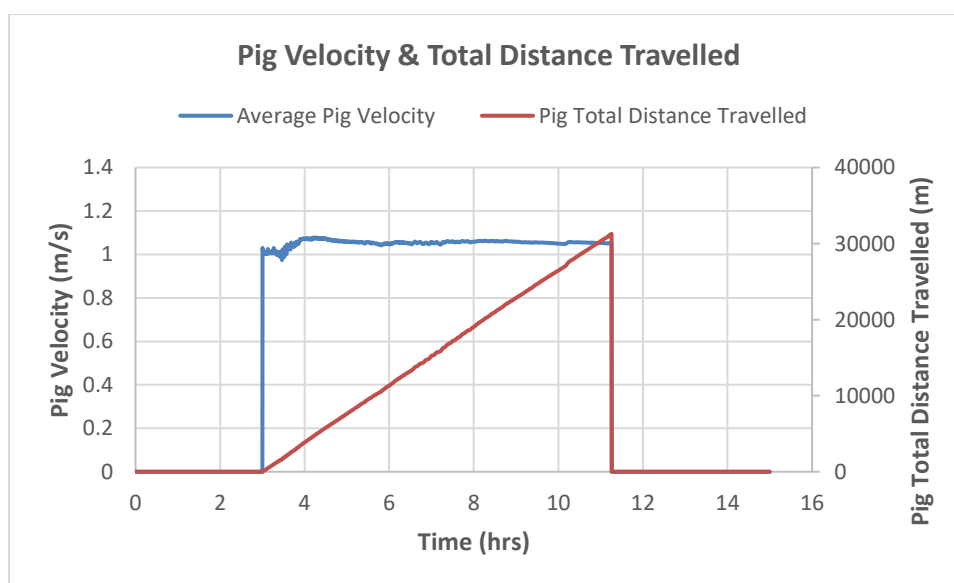


Figure 23 - Average Pig Velocity & Distance Travelled (1 m/s)

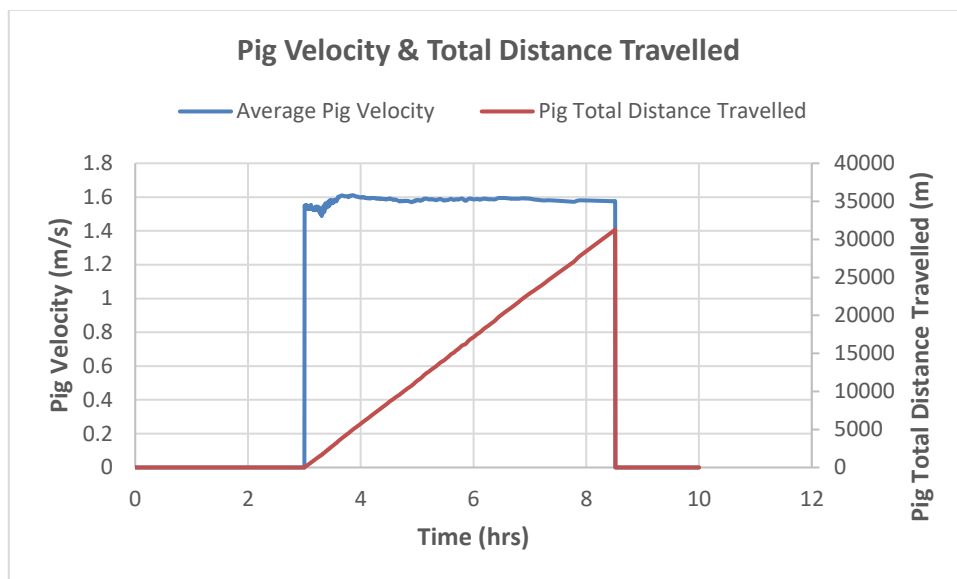


Figure 24 - Average Pig Velocity & Distance Travelled (1.5 m/s)

6.6.1. Liquid Content in Pipeline

Figures 25 and Figure 26 present the changes in the pipeline liquid contents as the pig travels the length of the pipeline

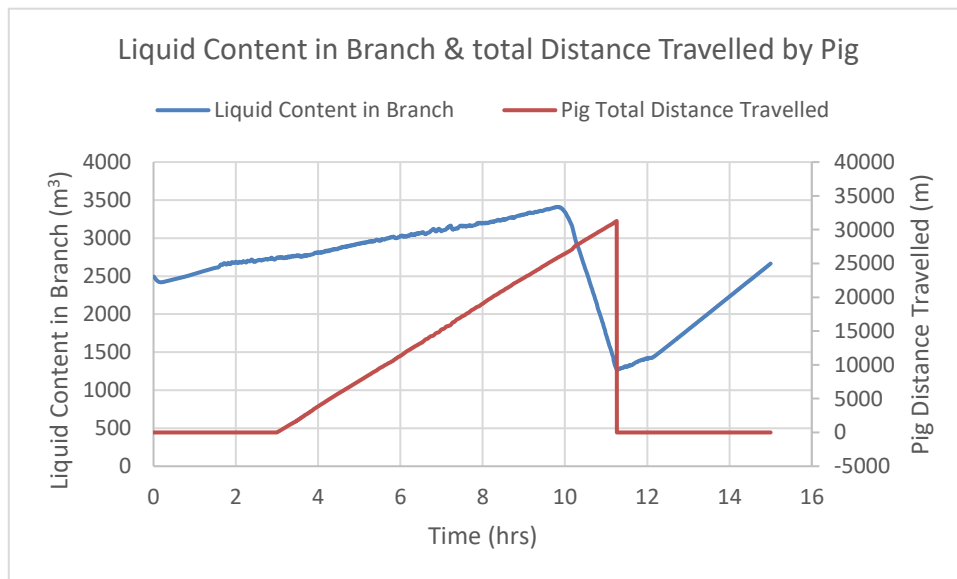


Figure 25 - Change in Pipeline Liquid Content with Pig Travel (1m/s)

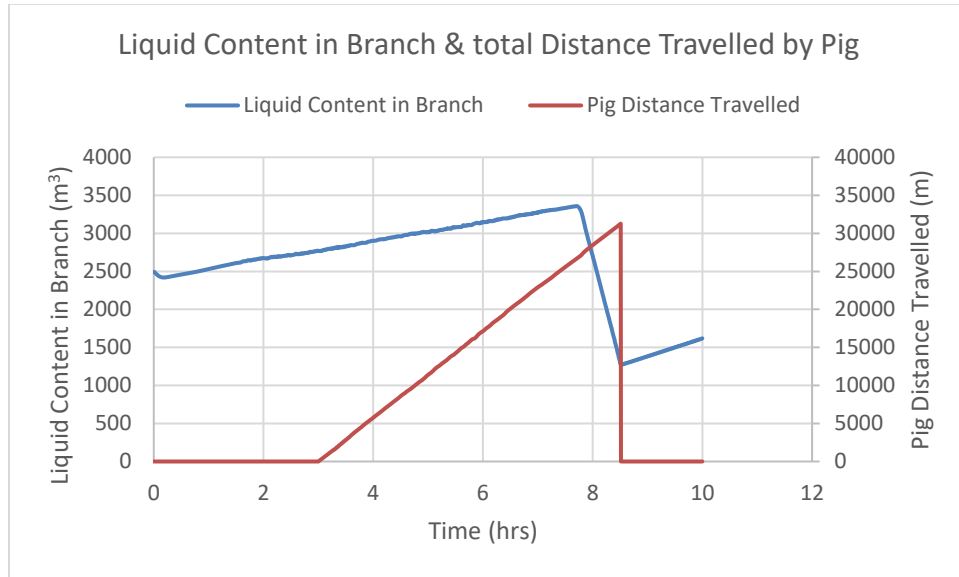


Figure 26 - Change in Pipeline Liquid Content with Pig Travel (1.5m/s)

6.6.2. Surge Volume During Pigging Operation

The surge volume is calculated as the integral of the difference between the instantaneous liquid flowrate entering the receiving vessel and a constant drain rate. The surge volumes as a function of drain rate are sum-up in the following table for a production line of 36 inch.

Table 11- Surge Volume as a function of flowrate (pig Velocity of 1.5m/s)

Drain Rate (m³/d)	Surge Volume (m³)
5000	2118
8000	2017
10000	1950
15000	1785
20000	1621
30000	1300
40000	989
45000	835
50000	682

Table 12 - Surge Volume as a function of flowrate (pig Velocity of 1 m/s)

Drain Rate (m ³ /d)	Surge Volume (m ³)
5000	2077
8000	1902
10000	1787
15000	1505
20000	1229
30000	717
40000	257
45000	51
50000	29